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Data Mining

Association Rule Mining on the Mushroom Dataset

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1. Introduction

One of the most widely used data mining techniques is association rule mining, which identifies relationships between variables by revealing how certain attributes tend to co-occur.

In this project, the Mushroom dataset from the OpenML repository `(ID: 24)` is analyzed. The dataset contains `8,124` instances representing mushroom specimens from the Agaricus and Lepiota families. Each instance is described by categorical attributes representing visual and structural characteristics, and is labeled as either edible (e) or poisonous (p).

The objective of this project is to apply association rule mining to the mushroom dataset in order to identify which mushroom characteristics are strongly associated with toxicity.

2. Dataset Description

2.1 Overview

The Mushroom dataset used in this project was obtained from the OpenML repository `(ID: 24)`. The dataset contains `8,124` instances, each described by categorical attributes representing physical characteristics, along with a class attribute indicating whether the mushroom is edible `(e)` or poisonous `(p)`. All features in the dataset are nominal (categorical) and encoded using single-letter symbols, allowing the dataset to be naturally transformed into a transactional format suitable for association rule mining algorithms such as Apriori. The key properties of the dataset are summarized in `Table 1`.

Dataset Property	***Value***
Dataset Name	Mushroom
OpenML ID	24
Source	UCI Machine Learning Repository (via OpenML)
Species / Family	Agaricus and Lepiota
Number of Instances	8,124
Number of Features	22 (+ 1 class attribute)
Feature Types	All nominal (categorical) - no numeric features
Class Attribute	class: edible (e) = 4,208 | poisonous (p) = 3,916
Missing Values	2,480 in stalk-root only (30.5% of instances)

Table 1. Mushroom Dataset — Key Properties

2.2 Features Description

`Table 2` lists sample of the Mushroom dataset attributes, along with their possible values.

Feature	*Possible Values*
cap-shape	bell(b), conical(c), convex(x), flat(f), knobbed(k),sunken(s)
bruises?	bruises(t), no(f)
odor	almond(a), anise(l), creosote(c), fishy(y), foul(f), musty(m), none(n), pungent(p), spicy(s)
stalk-shape	enlarging(e), tapering(t)
stalk-root	bulbous(b), club(c), cup(u), equal(e), rhizomorphs(z), rooted(r), missing(?)
veil-type	partial(p), universal(u)
class	edible (e) , poisonous (p)

Table 2. Sample Attribute Inventory and Possible Values

2.3 Class Distribution

The class attribute is nearly balanced, with **4,208** instances **(51.8%)** labeled as edible and **3,916** instances **(48.2%)** labeled as poisonous, as illustrated in **`Table 3`**. This balanced distribution prevents patterns from being dominated by a single majority class.

Class	*Instances*	*Percentage*
Edible (e)	4,208	51.8%
Poisonous (p)	3,916	48.2%

Table 3. Class Distribution (n = 8,124)

2.4 Attributes Cardinality

`Figure 1` shows the number of unique categories (cardinality) for each feature in the dataset. Cardinality ranges from 2 for binary attributes like bruises and stalk-shape to 12 for gill-color. Features with higher cardinality generate more items in the transaction matrix, increasing the number of potential combinations considered during association rule mining.

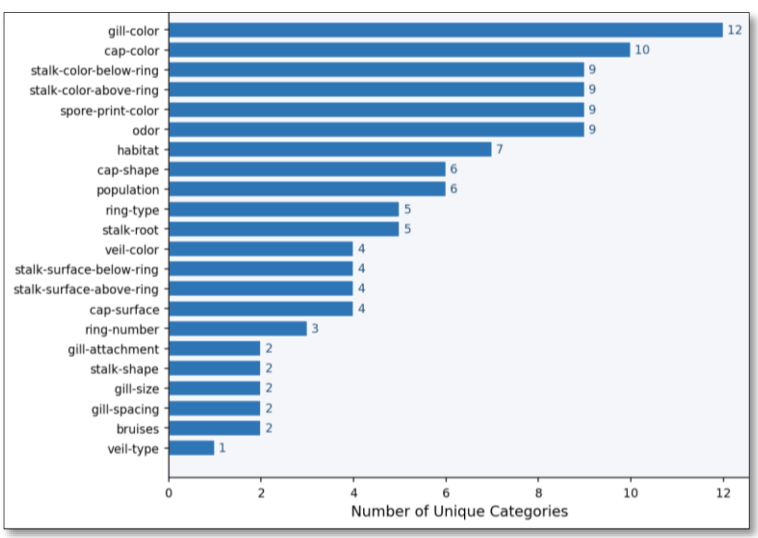


Figure 1. Attribute Cardinality — Number of Unique Categories per Feature

3. Data Preprocessing

Data preprocessing prepares the dataset for association rule mining. In this project, preprocessing was designed to preserve the relationships between categorical attributes while transforming the dataset into the binary transactional format required for association rule mining. The main steps include handling missing values, removing uninformative features, and encoding categorical attributes into a transaction matrix suitable for the **Apriori** algorithm.

3.1 Handling Missing Values

Inspection of the dataset shows that missing values occur only in the ***`stalk-root`*** attribute, encoded as **`"?"`**. As illustrated in **`Figure 2`**, **`2,480`** instances (**`30.5%`** of dataset) are affected.

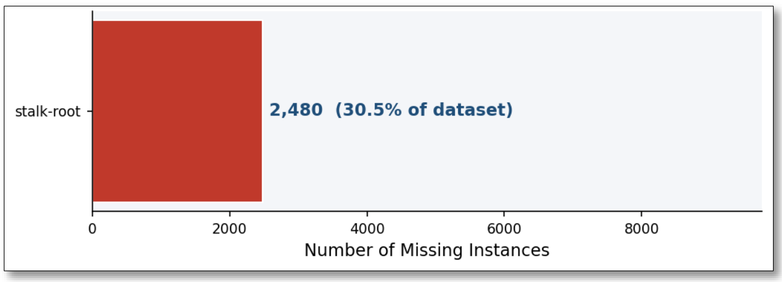


Figure 2. Missing Values in stalk-root — 2,480 instances (30.5% of dataset)

The handling approach was a two-step process, applied as follows in Python:

```
# Step 1: standardize the missing marker to NaN
df = df.replace('?', np.nan)

# Step 2: cast to string, then fill NaN with 'm' (missing category)
df = df.astype('string')
df = df.fillna('m')
```

Mode imputation was avoided because replacing missing values with the most frequent category would inflate its frequency and bias the mining results. Instead, missing entries were treated as a separate category **(`"m"`)**, allowing the algorithm to determine whether missing ***`stalk-root`*** values forms a meaningful pattern. The resulting distribution is shown in **`Figure 3`**, where the missing category (**`2,480`** instances) becomes the second largest group, confirming that removing the attribute or applying mode imputation would have significantly distorted the dataset.

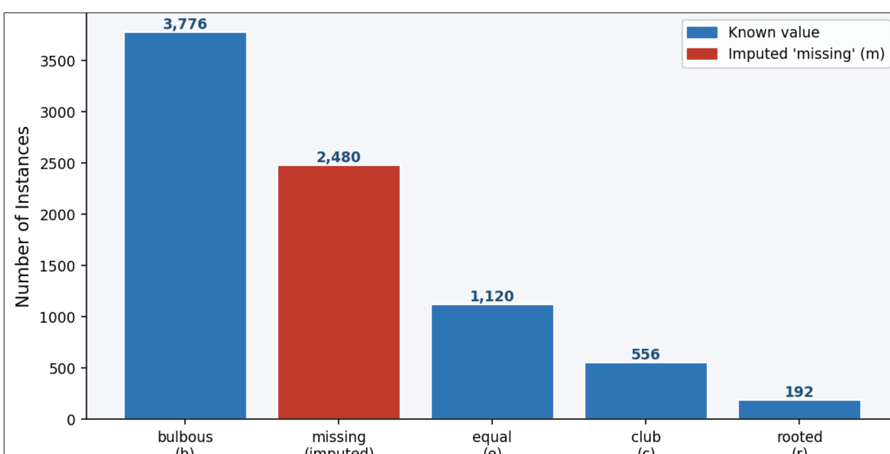


Figure 3. stalk-root Value Distribution After Missing Value Imputation

3.2 Constant Column Removal

During preprocessing, features were examined for zero variance. This check revealed that the attribute ***veil-type*** contains only one value, ***(p-partial)***, for all ***8,124*** mushrooms. Since this attribute provides no useful information for data mining, it was removed from the dataset, reducing the number of attributes from ***23 to 22*** prior to transaction encoding. The removal was implemented as follows:

```
nunique = df.nunique()

constant_cols = nunique[nunique <= 1].index.tolist()

if constant_cols:

    df = df.drop(columns=constant_cols)
# Result: constant_cols = ['veil-type']
```

3.3 Rare Category Analysis

Before encoding, category frequencies were analyzed using the minimum support threshold for rule mining ***(20%)***. Categories below this threshold are unlikely to appear as individual frequent itemsets. However, rare categories were retained to preserve the meaning of the attributes and allow potential participation in association rules when they co-occur with other features.

Table 4 lists the features containing at least one category below this threshold.

Feature	*Total Categories*	*Rare Categories*	*Rare Values*
gill-color	12	11	p, w, n, g, h, u, k, e, y, o, r
cap-color	10	8	e, y, w, b, p, c, u, r
stalk-color-above-ring	9	7	g, n, b, o, e, c, y
stalk-color-below-ring	9	7	g, n, b, o, e, c, y
odor	9	7	y, s, a, l, p, c, m
spore-print-color	9	5	r, u, o, y, b
habitat	7	5	p, l, u, m, w
population	6	4	s, n, a, c
cap-shape	6	4	k, b, s, c
stalk-root	5	3	e, c, r
ring-type	5	3	l, f, n
veil-color	4	3	n, o, y
ring-number	3	2	t, n
stalk-surface-above-ring	4	2	f, y
stalk-surface-below-ring	4	2	f, y
gill-spacing	2	1	w

cap-surface	4	1	g
gill-attachment	2	1	a

Table 4. Rare Category Analysis (Support Threshold = 20%)

3.4 Encoding for Association Rule Mining

Association rule mining algorithms operate on a binary transaction matrix, where each column represents an item and each row represents a transaction. The categorical dataset was converted into this format in two steps.

- **Contextual Labeling:** Each categorical value was prefixed with its attribute name to prevent ambiguity between identical symbols used by different features.

```
# Add contextual labeling:
for col in df_items.columns:
    df_items[col] = col + '=' + df_items[col]
# e.g. 'f' in odor becomes 'odor=foul'
#      'f' in cap-surface becomes 'cap-surface=fibrous'
```

- **Binary (One-Hot) Encoding:** The labeled dataset was then converted into a binary transaction matrix, where each column represents an attribute–value item and each cell indicates ***presence (1) or absence (0)***:

```
transaction_matrix = pd.get_dummies(
    df_items, prefix='', prefix_sep=''
).astype('int8')
# Result: 8,124 rows × 118 binary item columns
```

The resulting matrix contains ***8,124*** rows and ***118*** binary columns across the ***22*** retained features. ***Table 5*** summarizes the dataset dimensions across the preprocessing stages.

Stage	***Dimensions***	***Notes***
Raw dataset	8,124 × 23	23 categorical columns including class
Drop constant column	8,124 × 22	veil-type removed
Contextual labeling	8,124 × 22	Values become e.g. odor=foul
One-hot binary encoding	8,124 × 118	118 binary item columns

Table 5. Dataset at Each Encoding Stage

3.5 Normalization and Scaling

Scaling and normalization were not applied, as they are primarily designed for numerical, distance-based algorithms **(e.g., K-Means or SVM)**. In contrast, association rule mining operates on co-occurrence measures such as **support, confidence, and lift** computed directly from **binary** transaction data. Therefore, scaling would only introduce fractional values without meaningful interpretation and would not improve the quality of the resulting association rules.

4. Exploratory Data Analysis (EDA)

4.1 Summary statistics

4.1.1 Overview

Since the Mushroom dataset consists entirely of categorical attributes, traditional numerical statistics such as mean or standard deviation are not applicable. Instead, summary statistics are presented using frequency distributions to show how often each category appears in the dataset.

4.1.2 Summary of Feature Characteristics

Table 6 summarizes the number of categories per feature, the most frequent value, and whether the feature is binary. This combined table provides a concise overview of the dataset structure.

Feature	*No. of Categories*	*Most Frequent Category*	*count*	*Binary Feature?*
cap-shape	6	x	3656	F
cap-surface	4	y	3244	F
cap-color	10	n	2284	F
bruises?	2	f	4748	T
odor	9	n	3528	F
gill-attachment	2	f	7914	T
gill-spacing	2	c	6812	T
gill-size	2	b	5612	T
gill-color	12	b	1728	F
stalk-shape	2	t	4608	T
stalk-root	5	b	3776	F
stalk-surface-above-ring	4	s	5176	F
stalk-surface-below-ring	4	s	4936	F
stalk-color-above-ring	9	w	4464	F
stalk-color-below-ring	9	w	4384	F
veil-color	4	w	7924	F
ring-number	3	o	7488	F
ring-type	5	p	3968	F
spore-print-color	9	w	2388	F
population	6	v	4040	F
habitat	7	d	3148	F
class	2	e	4208	T

Table 6. Summary of Feature Characteristics

4.2 Visualizations and Interpretation of Patterns

To better understand the mushroom dataset and identify patterns related to toxicity, several visualizations were created using categorical plots. Since all attributes in the dataset are categorical, count plots and stacked bar charts were used instead of numerical plots such as scatter plots. These visualizations help reveal the distribution of important features and their relationship with the target variable ***`class`***.

4.2.1 Feature Distribution Across the Dataset

The first visualization shows the frequency distribution of the categorical attributes in the dataset. This provides an overview of how values are distributed across different features before analyzing their relationship with the target variable.

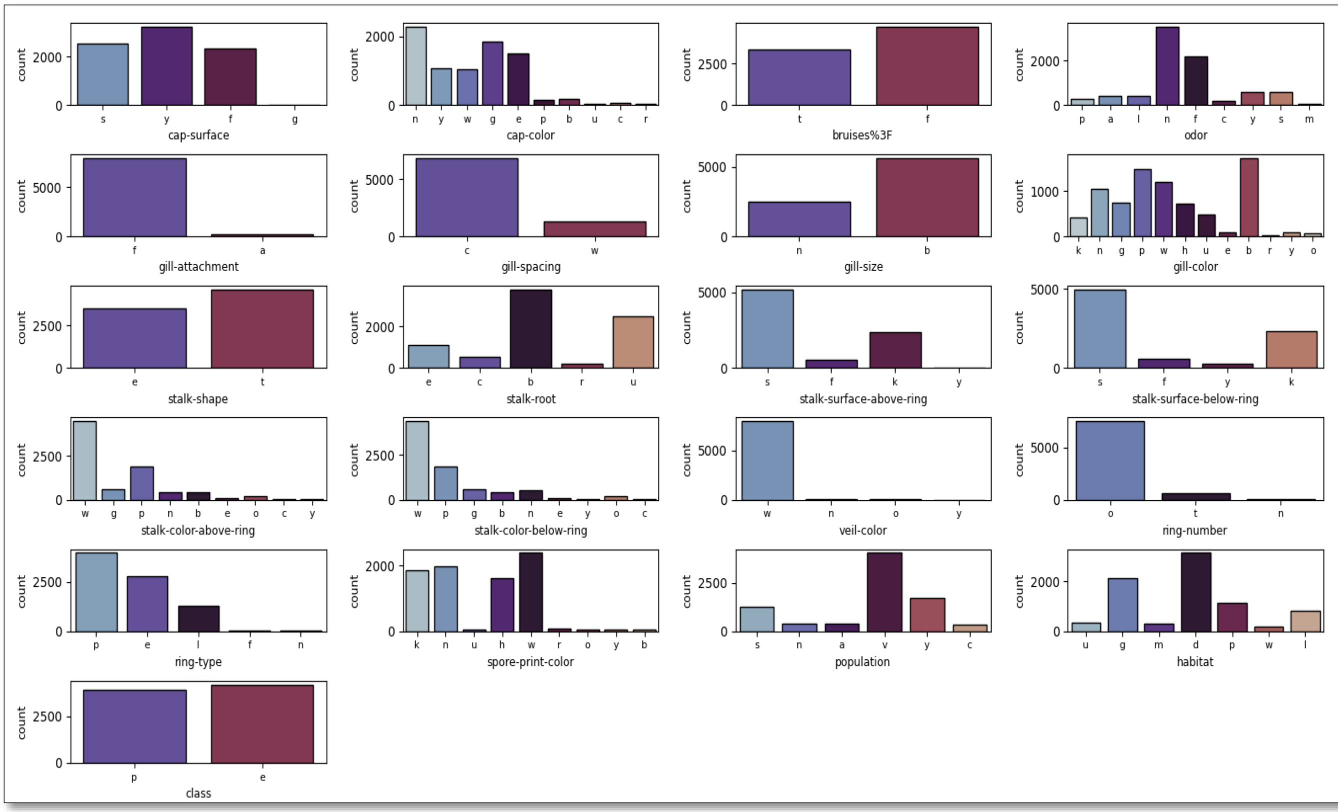


Figure 4. Distribution of All Categorical Features in the Mushroom Dataset

- **Interpretation:** Several attributes contain dominant categories, indicating that certain mushroom characteristics appear more frequently than others.

4.2.2 Class Distribution

The second visualization illustrates the distribution of mushroom classes in the dataset using a pie chart.

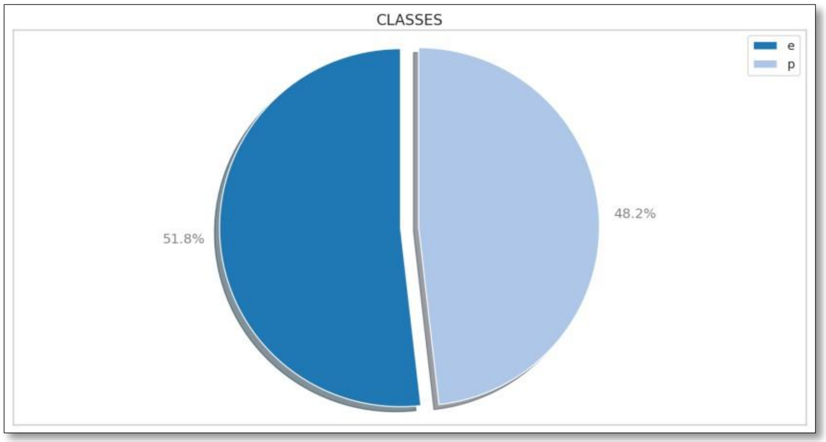


Figure 5. Distribution of Mushroom Classes

- **Interpretation:** The dataset does not suffer from severe class imbalance, as both edible and poisonous mushrooms are well represented. This balance is important because it helps reduce bias during later data mining or classification tasks.

4.2.3 Feature Relationships with Class

To explore the relationship between mushroom characteristics and toxicity, several count plots were generated comparing selected features with the target variable ***(class)***.

- **Odor vs. Class**

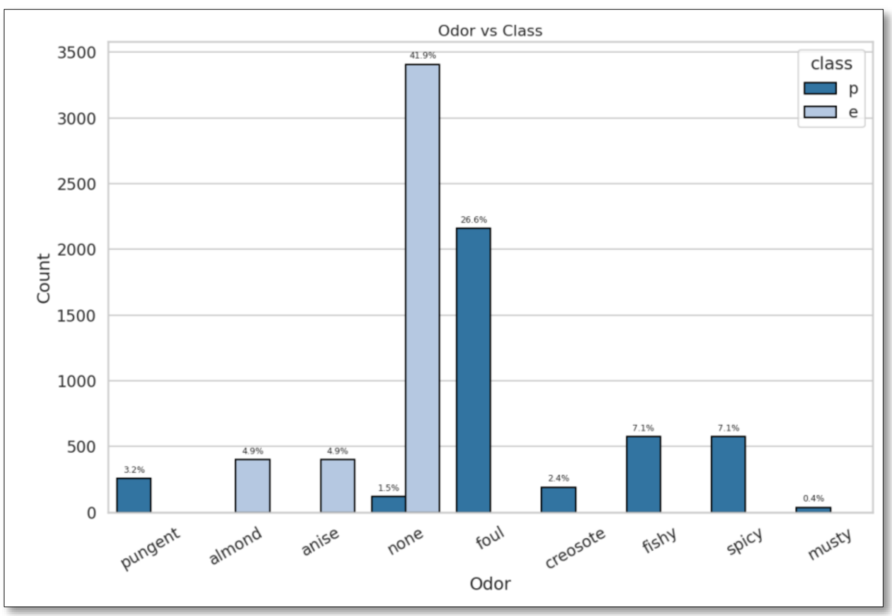


Figure 6. Odor Distribution by Mushroom Class

- **Interpretation:** ***Odor*** appears to be the strongest feature associated with mushroom toxicity, showing a clear separation between edible and poisonous mushrooms.

- **Gill Size vs. Class**

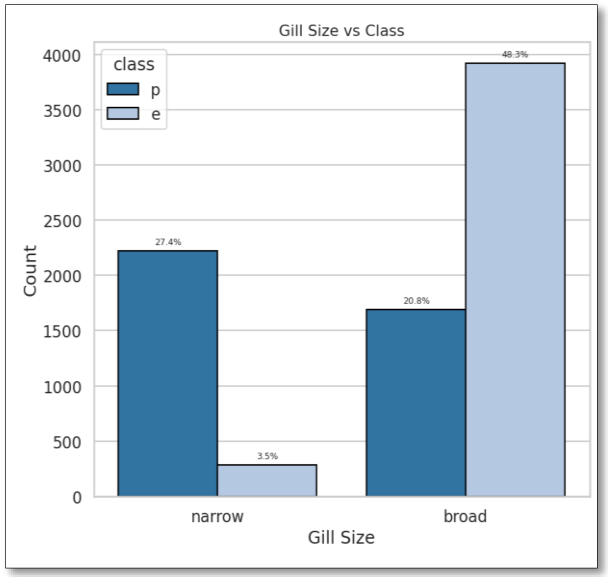


Figure 7. Gill Size Distribution by Mushroom Class

Interpretation: ***`Gill`*** **`Size`** shows a noticeable relationship with mushroom class and may help distinguish edible from poisonous mushrooms.

- **Stalk Surface Ring vs. Class**

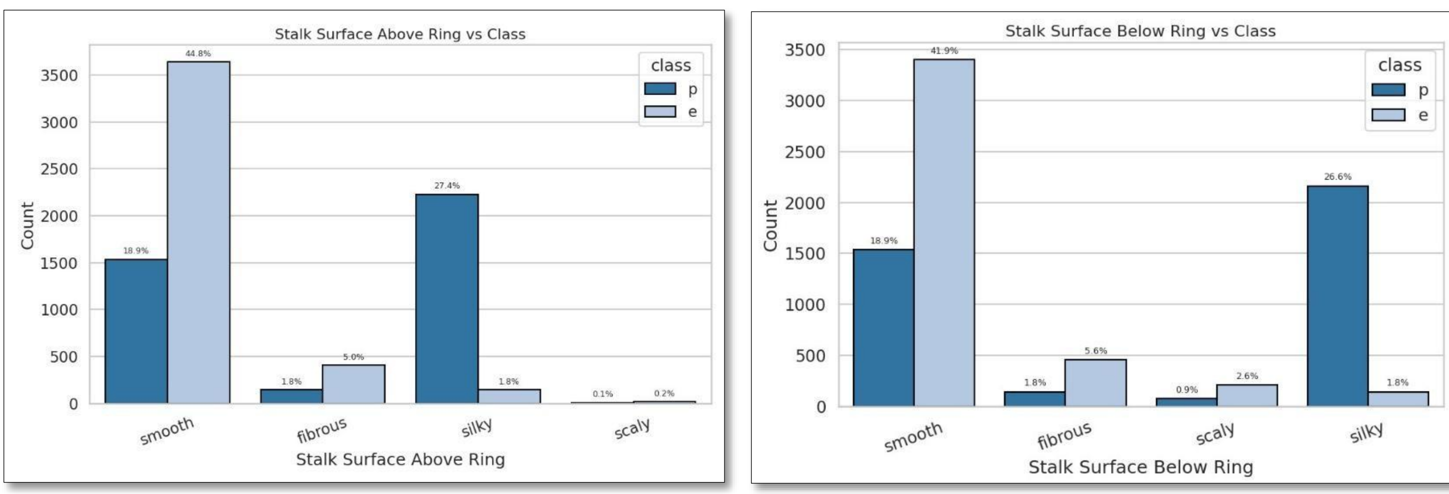


Figure 8. Stalk Surface Ring by Mushroom Class

Interpretation: Mushrooms with smooth **`stalk surfaces`** are more frequent in the **`edible`** class, while **`silky surfaces`** are more common in the **`poisonous`** class.

- **Spore Print Color vs. Class**

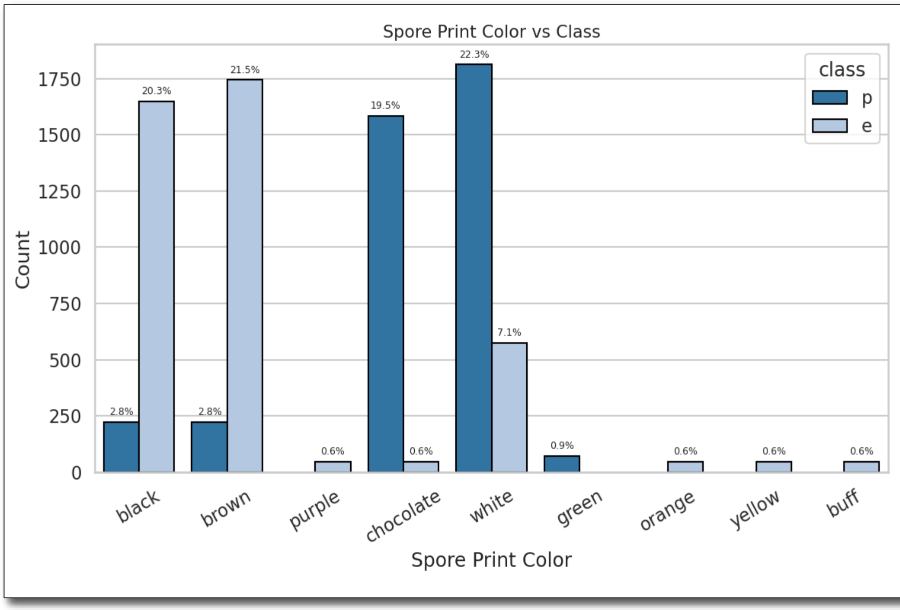


Figure 9. Spore Print Color Distribution by Mushroom Class

Interpretation: `Spore Print Color` shows noticeable variation between edible and poisonous mushrooms.

- **Gill Color vs. Class**

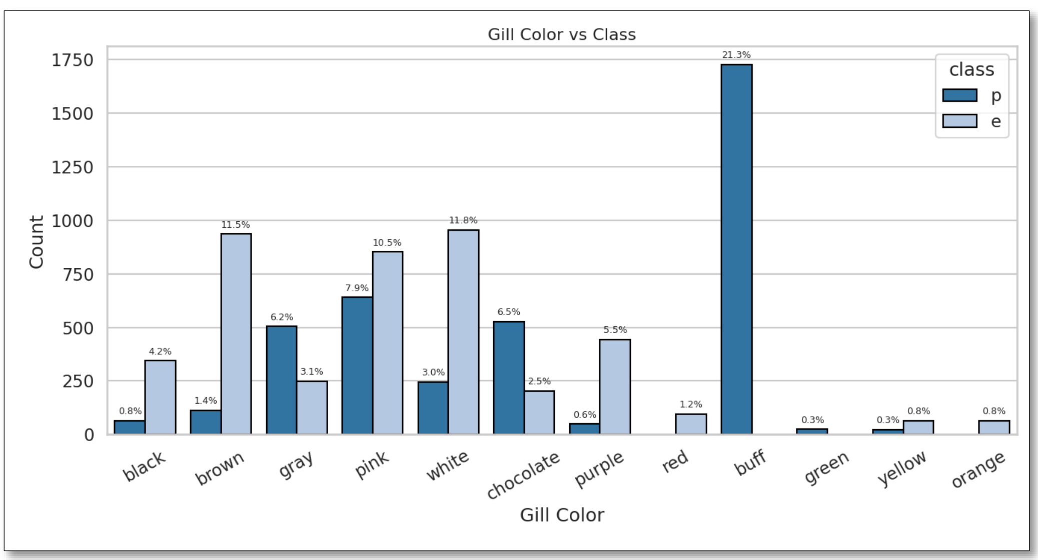


Figure 10. Gill Color Distribution by Mushroom Class

Interpretation: `Spore Print Color` shows noticeable variation between edible and poisonous mushrooms.

4.2.4 Feature Relationships with Class

From the visual analysis, several important observations were identified:

- The dataset is relatively balanced between edible and poisonous mushrooms.
- `Odor` is the strongest feature visually associated with mushroom toxicity.

- **`Gill size`** and **`Bruises`** also show clear relationships with the target class.
- **`Gill color`** and **`Spore Print Color`** provide additional useful patterns.
- **`Cap color`** is useful for describing the dataset, but it is less informative for distinguishing toxicity compared with other attributes.

Overall, the visualization stage provided a clearer understanding of the dataset and highlighted the most relevant attributes before proceeding to the data mining phase.

5. Data Mining Methodology

Association rule mining was applied using the **`Apriori`** algorithm to discover relationships between mushroom characteristics.

The dataset was converted into a transaction matrix where each row represents a mushroom instance and each column represents an item in the form **`attribute=value`**. The resulting matrix contained `8124` transactions and `118` items. The **`Apriori`** algorithm was applied with the following parameters:

- Minimum support: `0.20`
- Maximum itemset length: `2`

After extracting frequent itemsets, association rules were generated using a confidence threshold of `0.8`. The rules were evaluated using support, confidence, and lift metrics.

```
===== Association Rule Mining =====

Transaction Matrix Overview
(8124, 118)
   cap-shape=b  cap-shape=c  cap-shape=f  ...  habitat=w  class=e  class=p
0        False        False        False  ...      False    False     True
1        False        False        False  ...      False     True    False
2         True        False        False  ...      False     True    False
3        False        False        False  ...      False    False     True
4        False        False        False  ...      False     True    False

[5 rows x 118 columns]
```

Figure 11. Sample of The Binary Transaction Matrix

6. Results and Evaluation

The **`Apriori`** algorithm generated the following results:

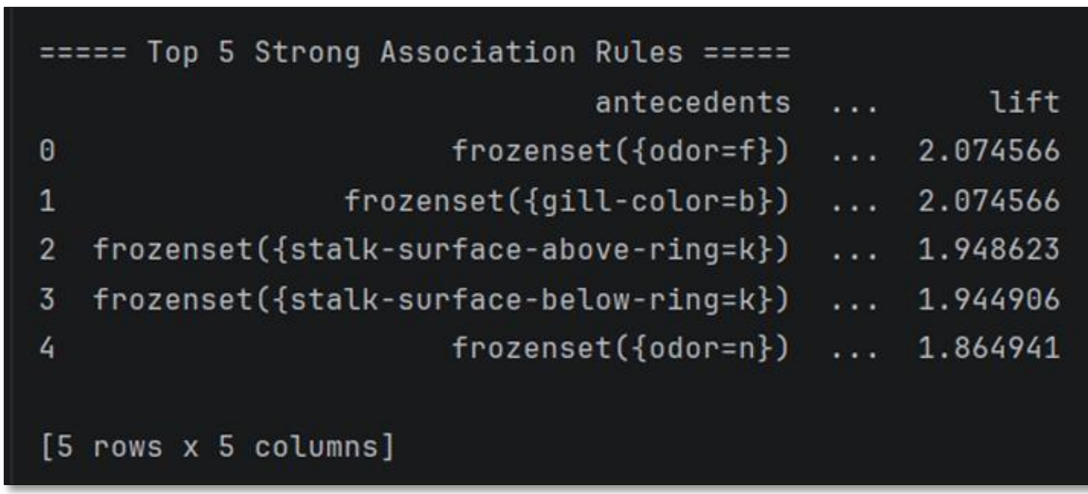
- `376` frequent itemsets
- `208` association rules
- `9` rules predicting the mushroom class

The strongest rules were selected based on confidence and lift values.

Rule	Support	Confidence	Lift
odor=f → class=p	0.2659	1.0000	2.0746
gill-color=b → class=p	0.2127	1.0000	2.0746
stalk-surface-above-ring=k → class=p	0.2742	0.9393	1.9486
stalk-surface-below -ring=k → class=p	0.2659	0.9375	1.9449
Odor=n → class=e	0.4195	0.9660	1.8649

Table 7. Top Association Rules Predicting Mushroom Class

The rule **odor=f → class=p** shows that mushrooms with a foul odor are always poisonous in the dataset. Similarly, **odor=n → class=e** indicates that mushrooms with no odor are highly likely to be edible. These rules demonstrate that odor and some structural mushroom features are strong indicators of toxicity.


```
===== Top 5 Strong Association Rules =====
                              antecedents  ...      lift
0                       frozenset({odor=f})  ...  2.074566
1                 frozenset({gill-color=b})  ...  2.074566
2  frozenset({stalk-surface-above-ring=k})  ...  1.948623
3  frozenset({stalk-surface-below-ring=k})  ...  1.944906
4                       frozenset({odor=n})  ...  1.864941

[5 rows x 5 columns]
```

Figure 12. Top 5 Strong Association Rules

7. Conclusion

Association rule mining successfully identified strong relationships between mushroom characteristics and their edibility. Several rules achieved very high confidence values, indicating reliable patterns in the dataset. However, these rules are derived from a specific dataset and may not fully generalize to real-world mushroom identification.